Hyper-**N**etwork for **e**lectro**Mo**bility

D3.3 NeMo Common Information Models

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Authors	Evangelia Portouli, Nikolaos Desypris (ICCS), Thomas Fousse (GIREVE), Jose Fernandez (MOSAIC), Roberto Tola (CRF), Joerg Kuefen (fKA), Nicolas Brailovsky (TOMTOM)
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List of abbreviations

Abbreviation	Meaning
CIM	Common Information Model, the models of objects and data structures as proposed by NeMo project
CPO	Charge Point Operator`
DoA	Description of Action
DSO	Distribution System Operator
EC	European Commission
eMAID	eMobility Account Identifier
eMI^3	eMobility ICT Interoperability Innovation Group
EMP or eMSP	ElectroMobility Service Provider
EV	Electric Vehicle
EVCO	Electric Vehicle Contract, contract between an EMP and a vehicle
EVSE	Electric Vehicle Supply Equipment
EVSP	Electric Vehicle Service Provider
ICT	Information and Communications Technologies
ISO	International Standards Organisation
MV	Medium Voltage
NFC	Near Field Communication
OTA	Over the Air
OWL-S	Ontology Web Language for describing Semantic Web Services
PLC	Programming Logic Control
PoD	Point of Delivery

Abbreviation	Meaning
RFID	Radio Frequency Identification
SoC	State of Charge (refers to vehicle battery)
SoH	State of Health (refers to vehicle battery)
VIN	Vehicle Information Number
WP	WorkPackage

Executive Summary

The NeMo project aims to develop a Hyper-Network of tools and services, to enable the interoperable and seamless service provision across countries and across service providers. One of the pillars of this Hyper-Network is the possibility to exchange data using a common format. The present document presents the Common Information Models proposed by NeMo for this scope. The vision is that the new services to be built using the Hyper-Network functionalities will generate and exchange data according the proposed Common Information Models, while data translators, to be developed in another NeMo task, will enable the translation of data from other formats to the NeMo Common Information Models.

The work was based on the data flows created within Task 2.1 of the project, to describe the sequence of information between electromobility actors per each of the seven Business Scenarios selected in WP1. Based on this work, the physical entities and the data structures that need modelling have been identified and the needed attributes have been agreed.

This Deliverable proposes models for all identified Business Objects, split in logical categories, as objects relating to the electric vehicle, to the charging infrastructure, to the final customer, to the charge session, to smart charging functionalities, to the market place for service creation and delivery, to grid loads due to the charging of electric vehicles and to the preparation of electric vehicles for drive-off. Supporting Business Objects are also modelled. For each attribute, the following are defined: Name, Definition, number of Instances (1 or n), whether it is a Must or Optional attribute (M/O) and its Format.

The proposed Common Information models will form the basis for the service creation and provision within the NeMo Hyper-Network while components will be proposed to translate data from proprietary formats to the NeMo language. It is envisaged that this common language will promote the interoperability of electromobility services and will enable the better user's experience. It is expected that during the project developments, some changes will be necessary in the proposals presented in this document. In such cases, the proposed models will be updated accordingly.

1. Introduction

The electromobility sector would benefit from the interoperability of relevant services offered, including charging and paying and finding and reserving an available charging station. Having recognised this need, the NeMo project aims to develop a Hyper-Network of tools and services, exactly to enable interoperable and seamless service provision across countries and across service providers. One of the pillars of this Hyper-Network is the possibility to exchange data using a common format. This is the rationale of Task 3.5 of the project, which is to define and propose Common Information Models for each electromobility object and data category. The vision is that the new services to be built using the Hyper-Network functionalities will generate and exchange data according the proposed Common Information Models, while data translators, to be developed in Task 3.2, will enable the translation of data from other formats to the NeMo Common Information Models. It is expected that this process will support the interoperability of electromobility services.

The present document is the outcome of Task 3.5 and presents the Common Information Models proposed by NeMo. Chapter 2 presents the methodology followed to define the models. Chapter 3 presents the models for physical objects which are relevant to an electric vehicle. Chapter 4 presents the models for data structures which are relevant to a charge session. Chapter 5 presents models of data structures which are relevant to smart charging functionalities. Chapter 6 presents models of data structures which are relevant to the market place under development in NeMo, while Chapter 7 includes models of data structures which are relevant to the final user of an electric vehicle. Chapter 8 presents models for data structures which are needed to create reports of local and regional grid loads due to the charging of electric vehicles. Chapter 9 presents models for data structures which are relevant to advice on the better preparation of electric vehicles, to achieve the highest possible battery capacity at drive-off. Chapter 10 presents models for physical objects relevant to the charging infrastructure, while Chapter 11 includes models for all supporting data structures, used by several of the main models presented in the previous chapters. Chapter 12 presents the conclusions of this work.

2. Methodology

The first phase of the work was to identify which are the objects and data structures that need to be modelled. This part of the work was based on the data flows created within Task 2.1 of the project, to describe the sequence of information between actors per Business Scenario.

In more details, NeMo has selected 64 use cases, synthesised in seven Business Scenarios, as the most relevant for the Hyper-Network under development. They are presented in the NeMo Deliverable 1.1, "Consolidated Version of the Use Cases and Actors' Requirements". Within Task 2.1, data flows have been created per Business Scenario. The data flows describe the information that is sent between the various physical entities, indicatively the customer, the electric vehicle, the EVSE, and the digital entities, indicatively the EMP or the CPO, sequentially as each Business Scenario evolves. They are presented in NeMo D2.1, "Consolidated Version of Technical Specifications for NeMo development and integration". Based on these data flows, the partners have identified the physical entities and the data structures that need modelling, to ensure interoperability of services using such data. The selected physical entities and data structures that are modelled in this Deliverable are denoted as Business Objects. At the same time, the attributes per Business Object have been agreed among partners, so that the description of each Business Object includes all necessary elements and information.

The Business Objects have been split in logical categories, as objects relating to the electric vehicle, to the charging infrastructure, to the final customer, to the charge session, to smart charging functionalities, to the market place for service creation and delivery, to grid loads due to the charging of electric vehicles and to the preparation of electric vehicles for drive-off. In order to describe the Business Objects, the template of the eMi^3 Electric Vehicle ICT Interface Specifications Part 2: Business Objects V1.0 was used. For each attribute, the following are defined: Name, Definition, number of Instances (1 or n), whether it is a Must or Optional attribute (M/O) and its Format. eMI³ is an open group of actors from the Electric Vehicles market who joined forces to harmonise the ICT data definitions, formats, interfaces, and exchange mechanisms in order to enable a common language among all ICT platforms for Electric Vehicles. Several NeMo partners are members of this group, so NeMo has based its work on these specifications. The eMI³ specifications describe the charging infrastructure business objects and the charge detail record business object only. NeMo is proposing models for all objects which are relevant for electromobility and is proposing updates to the eMI³ specifications according to the needs identified in its WP1 work. The updates are compared to the eMI³ specifications and are justified in the descriptions of the Charge Detail Record and of the charging infrastructure business objects.

The following data types and formats are used in the models based on common programming data type terminology:

Data type	Definition
int	32-bit signed two's complement integer, which has a minimum value of -231 and a maximum value of 231-1

long	64-bit two's complement integer, the signed long has a minimum value of -263 and a maximum value of 263-1
float	single-precision 32-bit IEEE 754 floating point
double	double-precision 64-bit IEEE 754 floating point
boolean	has only two possible values: true and false.
string	string of chars (char is a single 16-bit Unicode character, with a minimum value of '\u0000' (and a maximum value of '\uffff')
complex	model of another object in this document
list	list of a countable number of values
map	two-dimensional array, namely list of pairs
date	string according to ISO8601, i.e. YYYY-MM-DD
time	string according to ISO8601 UTC+Offset, i.e. hh:mm:ss [{+|-}hh:mm]
datetime	string according to ISO8601 UTC+Offset, i.e. YYYY-MM-DD hh:mm:ss [{+|-}hh:mm]

It must be noted that the object relations in the next chapters are sometimes described at a high level. A whole object may be included in the description of another object. Still, at low level implementation of the models, only the ID of the object may be included in the description of another object to provide the link to it.

3. Vehicle Business Objects

This section describes models of physical objects relevant to the electric vehicle.

Based on the Extended Vehicle approach of ISO 20078, it is envisaged that there will be a Neutral Server, which will separate the vehicle manufacturers' back-ends from the users and service providers. The vehicle manufacturers' back ends collect all real time data sent by the connected vehicles. A subset of these data can be provided to a service provider, after an agreement with a vehicle manufacturer. A subset of these data (for example VIN and location) can be sent to a service provider, only if an authorisation has been given by the vehicle owner.

3.1 Battery

This object models the battery pack available on an electric vehicle.

Each Battery object will be described by aggregating information coming from different sources, for example BatteryID, SoC, BatteryType and ChargingStatus will come from the Neutral Server. ChargeCompletionEstimatedTime may be calculated by a NeMo service.

ATTRIBUTE	DEFINITION	INST.	M/O	Format
BatteryID	Unique ID identifying the battery	1	M	string
VIN	The unique Vehicle Identification Number. It may be provided only after agreement by the vehicle owner.	1	O	string
BatteryType	The type of the battery. Possible values are: 0 – "Lead-acid" 1 – "Nickel metal hydride" 2 – "Zebra" 3 – "Lithium-ion" 4 – "Other"	1	O	int
BatteryCapacity	The battery capacity in kWh.	1	O	int
SoC	State Of Charge of the battery (%)	1	M	float
SoH	This is the ratio of the current battery capacity divided by the nominal capacity (in %). Typically, the battery capacity degrades with time.	1	O	double
BatteryFault	This indicates whether there is a generic battery fault or not. Possible values are:	1	O	int

	1 - “Fault” 2 - “No Fault”			
ChargeCompletionEstimatedTime	This is the estimated time (in m) until 100% charge of the battery.	1	O	int
ChargingStatus	This is the charging status of battery. Possible values are: 0 – “Cable connected and waiting, but not charging” 1 – “Charging” 2 – “Charge Interrupted” 3 – “Charge Complete”	1	O	int

3.2 Vehicle

This object describes an electric vehicle. It must be noted that there should be a previous agreement with the owner and service provider in order to allow the vehicle manufacturer to send data to the service provider according to the extended vehicle standard.

ATTRIBUTE	**DEFINITION**	**INST.**	**M/O**	**Format**
VIN	This is the unique Vehicle Identification Number	1	M	string
Manufacturer	The manufacturer of the vehicle	1	O	string
VehicleModel	The model of the vehicle.	1	O	string
ChargingMode	The charging mode of the charging cable available on the vehicle. It can take one of the following values: 1 – “Mode1-AC-1p” 2 – “Mode1-AC-3p” 3 – “Mode2-AC-1p” 4 – “Mode2-AC-3p” 5 – “Mode3-AC-3p” 6 – “Mode4-DC” 7 – “Legacy-Inductive”	n	M	List of int
MaxPower	The maximum amount of power (in W) that can be received during a charging session.	1	M	double

VehiclePower	The vehicle nominal power (in kW)	1	O	double
VehicleRange	The vehicle range based on average consumption (in km)	1	O	double
VehicleSpeed	The current vehicle speed (in km/h)	1	O	double
VehicleODO	The current indication of the vehicle odometer (in km)	1	O	int
VehicleConsumption	The vehicle average consumption (in kWh/100km)	1	O	double
VehicleRemainingDistance	The distance to destination (in km)	1	O	int
CurrentLocation	The current vehicle location	1	O	complex “LocationInfo”
DestinationLocation	The location of the vehicle destination	1	O	complex “LocationInfo”
ExternalTemperature	The external temperature (in C)	1	O	int
VehicleHeatingStatus	The status of the vehicle heating system. It can take the following values: 0 – “Heating off” 1 – “Heating on”	1	O	int
VehicleACStatus	The status of the vehicle AC system. It can take the following values: 0 – “AC off” 1 – “AC on”	1	O	int
PreconditioningFunctionalities	This is a list of the vehicle functionalities for which a notification for their pre-conditioning before drive-off may be sent. Possible values are: 0 – “Heating” 1 – “AC” 2 – “Ventilator” 3 – “Seat heating” 4 – “Windshield defrost”	n	O	list of int

	5 – “Odorant”			

4. Charge session Business Objects

This section describes models of data structures which have to be communicated between electromobility actors relevant to a charging session.

4.1 Authorisation

This business object models the information related to a user's authentication so that a charging session can start.

Authorisation refers to rules that determine who is allowed to do what, in contrast to authentication which is the process of ascertaining that somebody really is who he/she claims to be.

ATTRIBUTE	DEFINITION	INST.	M/O	Format
CPOSessionId	Session identifier, generated by the CPO, to identify the session/authorisation in the CPO's backend system.	1	M	string
eMSPSessionId	Session identifier, generated by the eMSP, to identify the session/authorisation in the eMSP's backend system. If the authorisation is offline, so done via whitelist of the eMSP on CPO or eRoaming Platform backend system, this information is empty.	1	O	string
sessionId	Main session identifier, generated by eRoaming Platform, to identify the session/authorisation. This ID is shared to all actors concerned by this session.	1	M	string
CPOID	This is the unique identifier of the CPO.	1	M	string
eMSPId	This is the unique identifier of the eMSP. If user uses an ad hoc payment mode, this value can be empty.	1	O	string
EVSEID	The identifier of the EVSE where the charging will take place	1	M	string
eMAID	This is the eMobility Account Identifier, as in ISO/IEC 15118. It refers to the account of an electromobility customer (with contract ID, service levels, etc.) and thus to any kind of an electromobility service that the service provider can offer.	1	M	string

ServiceId	This is the identifier of the service requested by this authorisation.	1	M	string
Status	The result of the authorisation check. It can take the following values: 1 – "Authorised" 0 - "Rejected"	1	M	int

4.2 ChargeDetailRecord

This business object models the information about a finished charge session. A charging session starts from the customer identification to any end of the session. One charging session consists of one or more charging periods.

ATTRIBUTE	DEFINITION	INST.	M/O	Format	Update of eMI[3] specifications
CPOSessionId	Session identifier, generated by the CPO, to identify the session/authorisation in the CPO's backend system.	1	M	String	New attribute
CDRID	This is unique identifier of the Charge Detail Record. According to the eMI[3] specifications it is unique per CPO and it consists of session ID AN36 (to allow unique random ID generation) and an instance corresponding to the current step in the CDR transmission chain (00 from CPO, 01 from first market place, 02 from 2nd market place)	1	M	string	
EVSEID	This is the unique identifier of the EVSE on which the charge was performed.	1	M	int	
eMAID	This is the customer's eMobility Account Identifier.	1	M	string	
EnergyMeterID	This is an identifier assigned by the EVSE operator, it can be the same as the PoDID.	1	O	string	
NumPhases	This indicates the type of power			integer	Replaces

	delivered. Possible values are: 0- “AC-single” 1- “AC-three” 2- “DC”				“Type of charge” attribute of the eMI[3] specifications
GlobalBeginSession	The start time of the charging session, it is equal to the start time of the first charging period.	1	M	datetime	
GlobalEndSession	The end time of the charging session, it is equal to the end time of the last charging period	1	M	datetime	
TotalEnergy	The total energy delivered to the vehicle during the whole charge session (in Wh).	1	M	float	
GlobalTransactionCost	The global cost of the transaction, including tax (in the currency specified below).	1	M	float	
Currency	The currency in which the cost is expressed, according to ISO 4217 numeric code	1	M	int	
ChargingPeriodRecords	The list of charging periods which comprise the global charging session	n	M	List of complex “ChargingPeriod Record”	

4.3 ChargingPeriodRecord

This business object models the information about one charging period. One charging session consists of one or more charging periods.

ATTRIBUTE	DEFINITION	INST.	M/O.	Format
Start_datetime	Start date and time of the charging period. The event is decided by the CPO, for example log on with badge, plug in …	1	M	datetime
End_datetime	End date and time of the charging period. The triggering event is decided by the CPO, for instance charging power going down to 0kW.	1	M	datetime
Type	The type of the charging period. Possible	1	M	int

	values are: 0 - “Occupied” 1 - “Charging” 2 - “Discharging”			
MaxPowerAllowed	The maximum power allowed during the charging period (in kW)	1	M	int
Duration	Duration of the charging period	1	M	string [hh]:[mm]:[ss]
Energy	The energy delivered to the vehicle during the charging period (in Wh).	1	M	double
DetailedCost	The cost of the transaction, including all cost like energy cost, parking cost, service fee, taxes (in the currency specified below).	1	M	float
Currency	The currency in which the cost is expressed, according to ISO 4217 numeric code	1	M	int

5. Smart Charge Business Objects

This section describes models of data structures which have to be communicated between electromobility actors in order to offer smart charging services, as planned in NeMo Business Scenarios 1, 2 and 3.

5.1 UserMobilityNeed

This Business Object describes the data structure that is sent from an electromobility service provider (EMP) or from a customer's device, in order to schedule a charging session for an electric vehicle before the next trip.

ATTRIBUTE	DEFINITION	INST.	M/O	Format
eMAID	This is the eMobility Account Identifier, as in ISO/IEC 15118. It refers to the account of an electromobility customer (with contract ID, service levels, etc.) and thus to any kind of an electromobility service that the service provider can offer.	1	M	string
RequestID	This is a unique identifier of the request	1	M	string
Trip	This is the description of the planned next trip	1	M	complex "Trip"
Vehicle	This is a reference to the description of the vehicle.	1	O	complex "Vehicle"
VehicleLoad	This is the expected vehicle load (in kg) during the next trip.	1	O	double

5.2 UserChargeNeed

This Business Object describes the data structure that is calculated and includes the energy needs for the customer's vehicle to perform the trip planned in the customer's request.

ATTRIBUTE	DEFINITION	INST.	M/O	Format
eMAID	This is the eMobility Account Identifier, as in ISO/IEC 15118. It refers to the account of an electromobility customer (with contract ID, service levels, etc.) and thus to any kind of an electromobility service that the service provider can offer.	1	M	string
RequestID	This is a unique identifier of the	1	M	string

	request			
ItinerarySections	The list of itinerary sections that comprise the next trip	n	M	List of complex “ItinerarySection”
Vehicle	This is a reference to the description of the vehicle.	1	O	complex “Vehicle”
ItineraryEnergyNeed	This is a list of the energy need (in kWh) of the vehicle at the start of each itinerary section in order to complete the next itinerary section	n	M	List of double
TripEnergyNeed	This is the total energy need (in kWh) of the vehicle in order to complete the next trip	1	M	double

5.3 ChargeProposition

This Business Object describes the data structure which proposes a list of EVSEs and offered charging profiles that can cover the customer’s request.

ATTRIBUTE	DEFINITION	INST.	M/O	Format
eMAID	This is the eMobility Account Identifier, as in ISO/IEC 15118. It refers to the account of an electromobility customer (with contract ID, service levels, etc.) and thus to any kind of an electromobility service that the service provider can offer.	1	M	string
RequestID	This is a unique identifier of the request	1	M	string
ChargeNeed	This is the reference to the structure with the information which was used as basis to result in the charge propositions	1	M	complex “UserChargeNeed”
ChargePropositionDetail	This is a list of charging propositions offered. There may be more propositions, even for the same EVSE	n	M	List of complex “ChargePropositionDetail”

5.4 ChargePropositionDetail

This Business Object models the information about one offered charging profile and cost. There is one business object per charging profile offered, even for the same EVSE.

ATTRIBUTE	DEFINITION	INST.	M/O	Format
eMAID	This is the eMobility Account Identifier, as in ISO/IEC 15118. It refers to the account of an electromobility customer (with contract ID, service levels, etc.) and thus to any kind of an electromobility service that the service provider can offer.	1	M	string
RequestID	This is a unique identifier of the request	1	M	string
ChargePropositionID	This is a unique identifier for the proposition, including the EVSE and the charging profile	1	M	string
ChargingPool	The pool offering the proposition	1	M	complex “ChargingPool”
EVSE	The EVSE offering the proposition	1	M	complex “EVSE”
ChargingProfileOffer	The charging profile offered in this proposition	1	M	complex “ChargingProfile”
GlobalCost	The global cost of the proposition, including all taxes (in the currency specified below).	1	M	float
Currency	The currency in which the cost is expressed, according to ISO 4217 numeric code	1	M	int
Tariff	This is the price per kWh in the currency used. It is calculated as GlobalCost / Energy	1	O	float
Discount	This is the discount (in %) offered by the CPO for the specific proposition, if any.	1	O	float
BookingFee	The cost for booking the proposition, (in the currency specified).	1	O	float
CancellationFee	The cost for cancellation (if any) of the proposition if accepted (in the currency specified).	1	O	float

5.5 PropositionReservationRequest

This Business Object models the data structure sent in order to book an offered charging proposition.

ATTRIBUTE	DEFINITION	INST.	M/O	Format
eMAID	This is the eMobility Account Identifier, as in ISO/IEC 15118. It refers to the account of an electromobility customer (with contract ID, service levels, etc.) and thus to any kind of an electromobility service that the service provider can offer.	1	M	string
RequestID	This is a unique identifier of the request	1	M	string
ChargePropositionID	The unique identifier of the selected charging proposition	1	M	string
ReservationStartEnd	The estimated date and time of arrival and departure at the selected EVSE	1	M	complex “TimePeriod”

5.6 VariableOffers

This Business Objects models information relevant to the electricity grid, as regards the power available, the maximum energy that can be supplied and the prices for charging per day and time in a specific area. Such information can be based on Dispatching Markets price signals and offers and/or DSO local constraints regarding medium voltage dispatching or network unbalances. It is expected that both needs will be dealt with between an Energy Retailer/Aggregator and CPOs or EMPs or between a DSO and CPOs or EMPs and the results of power and energy availability and charging costs for each area will be available via the NeMo Hyper-Network.

An area is mapped by a polyline, defined by n nodes.

ATTRIBUTE	DEFINITION	INST.	M/O	Format
AreaNodeGeocoordinates	This is a list of the geo-coordinates of each of the n nodes of the polyline defining the area.	n	M	List of complex “GeoCoordinates”
OfferValidFromTo	This is the start and end date and time of the offer validity	1	M	complex “TimePeriod”
PowerOffer	List of the maximum power offered in	n	O	List of

	the area with time			complex “PowerTime”
EnergyOffer	List of the energy availability in the area with time	n	O	List of complex “EnergyTime”
CostOffer	List of the cost per kWh in the area per time	n	O	List of complex “CostOffer”
LastUpdate	The date and time on which the offer description has been last updated	1	O	datetime

6. Marketplace Business Objects

This section describes models of the data structures, which are mainly relevant for the NeMo Business Scenario 4, namely for the marketplace for electromobility services creation and delivery.

6.1 ServiceContract

This Business Object models a contract signed between two entities.

ATTRIBUTE	DEFINITION	INST.	M/O	Format
ContractId	This is a unique identifier of the contract	1	M	string
Status	This is the status of the service contract. Possible values are: 1 – "Active" 2 – "Inactive" 3 – "Suspended"	1	M	int
CreationDate	This is the timestamp of the contract creation	1	M	datetime
ValidFromThrough	This is the time period from start to the end date of validity of the contract.	1	M	complex "TimePeriod"
LastChange	This is the timestamp of the last change of the contract.	1	O	datetime
ContractSection	This is a list of the text sections of the contract	n	M	List of complex "ContractSection"
Service	This is a reference to the list of services provided by the contract	n	M	List of complex "Service"
BusinessPartner	This is a reference to the list of Business Partner who contracts the service	1	M	complex "BusinessPartner"

6.2 ContractSection

This Business Object describes one section of a Contract. It is an intermediate object between the Contract and the Terms which needs to have a type. One possible use of this object is for display purposes.

ATTRIBUTE	DEFINITION	INST.	M/O	Format
ContractId	This is a unique identifier of the contract	1	M	string
SectionId	This is a unique identifier of the contract section	1	M	string
Name	Name of the contract section	1	M	string
Description	This is a description of the contract section	1	M	string
Type	This is the type of the contract section. Possible values are: 1 – “Descriptive” 2 – “Defining”	1	M	int
Terms	This is a list of the contact terms in the section	n	M	List of complex “Terms”

6.3 Terms

This Business Objects describes one term of a Contract.

ATTRIBUTE	DEFINITION	INST.	M/O	Format
ContractId	This is a unique identifier of the contract	1	M	string
SectionId	This is a unique identifier of the contract section	1	M	string
TermId	This is a unique identifier of the contract term or condition	1	M	String
Name	This is the name of the contract term	1	O	String
Type	Type of the contract term Possible values are: 1 – “Descriptive” 2 – “Defining”	1	M	int
Description	This is the description / content of the contract term	1	M	String
valueString	This is for a value used for computing the actual consumption of the service, for example “price”	1	O	string
valueDouble	This is the actual value of the valueString attribute	1	O	double

6.4 BusinessPartner

This Business Object describes a business partner.

ATTRIBUTE	DEFINITION	INST.	M/O	Format
PartnerID	This is a unique identifier of the business partner.	1	M	string
AdditionalIdD	These are additional identifiers relevant to the business partners, for example the eMAID	n	O	List of complex “AdditionalID”
dn	This is the distinguished name of the business partner, Typically used with an ldap server to centrally manage the entities, for example: (Ex☹cn=nemo,o=ibm,cn=ibmresearch)	1	O	string
CompanyName	This is the legal name of the business partner	1	M	string
CompanyAcronym	This is the acronym of the company's name	1	O	string
LegalRegistration	This is the company legal registration number	1	O	string
Status	This is the status of the business partner. Possible values are: 1 – “Pending” 2 –“Active” 3 – “Inactive” 4 – “Suspended”	1	M	int
BusinessPartnerAddress	This is a list of addresses of the business partner	n	M	List of complex “AddressInfo”
BusinessPartnerInformation	Additional business partner information	1	O	complex “BusinessPartnerInformation”
BusinessPartnerPersons	These are people from the business partner organisation who are users of the NeMo Hyper-Network	n	O	List of complex “Contact”

LastUpdate	The timestamp on which the business partner description has been last updated	1	O	datetime

6.5 BusinessPartnerInformation

This Business Object includes additional information about a business partner.

ATTRIBUTE	DEFINITION	INST.	M/O	Format
companyDescription	Description of the company	1	O	string
companyURL	URL of the company website	1	O	string
companyType	Type of the company. Possible values are: “Retailer” “OEM” “Regulator”	1	O	string
noEmployee	Number of employees of the company. Possible values are: 1 – “1-10” 2 – “10-100” 3 – “100-1000” 4 – “1000-5000” 5 – “>5000”	1	O	int

6.6 AdditionalID

This Business Object models additional identifiers which are connected to the Business Partner, for example the eMAID.

ATTRIBUTE	DEFINITION	INST.	M/O	Format
idType	This is the type of the identifier. It can have the following values: 1 – “eMAID” 2 – EVCOID 3 – “EVSEID”	1	M	int
idValue	This is the actual value of the identifier	1	M	string

6.7 Service

This Business Object describes a Service. The semantic service description is not part of the core NeMo Business Object Model - but already exists within an OWL-S description that is referenced from the object.

ATTRIBUTE	DEFINITION	INST.	M/O	Format
ServiceId	This is a unique identifier of the service	1	M	string
Name	This is a short description of the service	1	M	string
Description	This is a long description of the service	1	M	string
Category	This is a list of the categories of the service, for example charging, billing, etc.)	n	M	List of complex “Category”
BusinessPartner	This is a reference to the business partner offering the service	1	M	complex “BusinessPartner”
Status	This is the status of the service. Possible values are: 1 – “Draft” 2 – “Pending” 3 – “Active” 4 – “Inactive” 5 – “Suspended”	1	M	int
webService	This defines whether the service is a typical web service (like WSDL, REST) or is it an archive that can be downloaded and embedded	1	M	boolean
Specification	This is the url reference to the service specification (WSDL , REST, etc.)	1	M	string
TestInstructions	This is the url reference to information for testing the service.	1	O	string
SemanticDescription	This is a url reference to a Semantic Service Description. This is not a NeMo Business Object, but an existing standard (e.g. OWL-S)	1	O	string
Assets	This is a url reference to coded and/or non coded artefacts for the service	1	O	string

6.8 Category

This Business Object describes the service category taxonomy. It is a hierarchy allowing the Hyper-Network actors' to navigate through the service catalogue.

ATTRIBUTE	DEFINITION	INST.	M/O	Format
categoryId	This is an identifier of the service category	1	M	string
Name	This is the name of the service category	1	M	string
parentCategory	This is the parent of the category	1	O	complex "Category"
subcategory	This is a list of the subcategories of the category	n	O	List of complex "Category"

6.9 ServiceContractOffering

This Business Object aggregates information about a service and the terms and condition under which any service in the NeMo Hyper-Network may by contracted.

ATTRIBUTE	DEFINITION	INST.	M/O	Format
offeringId	This is a unique identifier of the service contract offering	1	M	string
offeringDate	This is a timestamp of the last offering change.	1	M	datetime
validFromThrough	This is the time period from the start to the end date of the offering validity	1	M	complex "TimePeriod"
Status	Status of the contract offering. Possible values: 1 – "Draft" 2 – "Pending" 3 – "Active" 4 – "Inactive" 5 – "Suspended"	1	M	int
ContractSection	List of text sections of the contract offering	n	M	List of complex "ContractSection"
Service	This is a reference to service provided by the offering	1	M	complex "Service"

7. User Business Objects

This section includes Business Objects describing a physical person who is using an EV and his/her preferences. These objects are mainly relevant for Business Scenarios 5, 1, 2 and 3.

7.1 User

This Business Object models the user of an EV and of electromobility services.

ATTRIBUTE	DEFINITION	INST.	M/O	Format
eMAID	This is the eMobility Account Identifier, as in ISO/IEC 15118. It refers to the account of an electromobility customer (with contract ID, service levels, etc.) and thus to any kind of an electromobility service that the service provider can offer.	1	M	string
ContactInformation	This is the contact information for the user	1	M	complex “Contact”

7.2 UserProfile

This business object models the preferences, history and agenda of the user.

ATTRIBUTE	DEFINITION	INST.	M/O	Format
User	This is the reference to the Customer	1	M	complex “User”
ChargingPreferences	These are the user’s charging preferences	1	M	complex “UserChargingPreferences”
DrivingPreferences	These are the user’s driving preferences	1	O	complex “UserDrivingPreferences”
TripHistory	This is a list of the registered trips of the User	n	M	List of complex ”Trip”
RecurrentPlaces	This is a list of the places frequently visited by the User	n	M	List of complex "Recurrent Place”
RecurrentTrips	This is a list of trips recurrently done by the User	n	M	List of complex "Recurrent Trip”

7.3 UserChargingPreferences

This Business Object models the charging preferences of a user.

ATTRIBUTE	DEFINITION	INST.	M/O	Format
User	This is the reference to the User	1	M	complex "User"
RangeAnxiety	This is the vehicle range (in km) under which the user feels uncomfortable	1	O	double
PreferredPools	This is the list of the charging pools preferred by the user	n	O	List of complex "ChargingPool"
PreferredCPOs	This is a list of the charge point operators preferred by the user	n	O	List of complex "CPO"
PreferredChargeLocations	This is a list of the charging locations preferred by the user	n	O	List of complex "LocationInfo"
PreferredTimeSlow	This is a list with the user's preferred time periods for slow charging	n	O	List of complex "TimePeriod"
PreferredTimeFast	This is a list with the user's preferred time periods for fast charging.	n	O	List of complex "TimePeriod"
PreferredEVSEPower	The user's preference as regards the rated power level of an EVSE (in W).	1	O	double
PreferredTarrif	The user's preference as regards the maximum kWh cost to pay for a charge session, including electricity, parking, taxes (in the currency provided).	1	O	float
Currency	The currency in which the cost is expressed, according to ISO 4217 numeric code	1	M	int
PreferredAuthorisation	The user's preferences as regards authorisation method. Possible values	n	O	int

	are: 1 - "Free" 2- "Private" 3 - "RFID-MIFARE" 4 - "RFID-DESFIRE" 5 - "RFID-Calypso" 6 - "PINPad" 7- "Phone-App" 8 - "Phone-NFC" 9 - "15118-PLC" 10 - "15118-OTA" 11 – “Phone” 12 – “Phone-SMS” 13 - "Credit-Card" 14 - "Cash" 15- "Pre-paid"			
PreferredCPFeatures	This is a list of preferences relevant to the charging infrastructure features. Possible values are: 1 – “Only renewable energy” 2 – “Free parking” 3 – “Bookable”	n	O	List of int

7.4 UserDrivingPreferences

This Business Object models the driving preferences of an EV user.

ATTRIBUTE	DEFINITION	INST.	M/O	Format
User	This is a reference to the User	1	M	complex “User”
AverageConsumption	Average Consumption of the driver based on his/her historical data (in kW / km)	1	O	double
ConsumptionPattern	The user’s preferred consumption pattern. It can take the following values: 1 – “Energy Saving” 2 – “Moderate”	1	O	int

	3 – “Sportive “			

7.5 UserAgenda

This Business Object models the agenda of the user that contains his/her next appointments and their locations.

ATTRIBUTE	DEFINITION	INST.	M/O	FORMAT
User	This is a reference to the User	1	M	complex “User”
Appointments	List of the appointments of the user filtered by date	n	O	Map of <datetime, Appointment>

8. eMobility Reporting Business Objects

This section describes models of the data structures, which are relevant for Business Scenario 6, namely for creating grid load reports and forecasts. It is envisaged that the charging stations include HW and SW to measure the energy delivered to each customer and charge the appropriate cost.

A DSO fiscal smart meter can be installed at several locations, for example at a single isolated charging station or at a charging pool. The measurements of the DSO fiscal smart meter are used by the Energy Retailers to charge the energy used, on a specific time basis, usually monthly or bimonthly.

8.1 GetLoadReport

This Business Object models the data structure that is sent by an authority or an energy retailer, in order to get the list of Charge Detail Records and the energy delivered per EVSE for a specific time period.

ATTRIBUTE	DEFINITION	INST.	M/O	Format
EnergyRetailerID or AuthorityID	This is a unique identifier of the energy retailer or of the authority that initiates the request	1	M	string
RequestID	This is a unique identifier of the request	1	M	string
CPO	This is a list of the CPOs from whom the data are requested	n	M	List of complex “CPO”
FromTo	Start and end time of the time period for which data are requested	1	M	complex “TimePeriod”

8.2 CPOLoadReport

This Business Object models the data sent by a CPO including the Charge Detail Records issued and the energy delivered per EVSE at the CPO’s infrastructure.

ATTRIBUTE	DEFINITION	INST.	M/O	Format
RequestID	This is a unique identifier of the request, to which the CPO responds	1	M	string
CPO	The CPO who provides the data	1	M	complex “CPO”
FromTo	Start and end time of the time period for which data are requested	1	M	complex “TimePeriod”

LoadDetails	List of the data provided per each EVSE	n	M	List of complex “LoadDetails”
CumulativeEnergy	This is the total energy delivered by the CPO in the time period requested (in kWh)	1	M	double

8.3 LoadDetails

This Business Object models the data sent by a CPO as regards the Charge Detail Records generated and the energy delivered per EVSE for a specific time period.

ATTRIBUTE	DEFINITION	INST.	M/O	Format
EVSEID	The unique identifier of the EVSE for which data is provided.	1	M	string
LocationInfo	This is the location of the EVSE	1	**O**	complex “LocationInfo”
CDRList	This is a list of CDRs generated by the EVSE in the requested time interval	n	M	List of complex “ChargeDetailRecord”
CPEnergy	The total energy delivered by this EVSE for the requested time interval (in kWh).	1	M	double

8.4 AreaLoadReport

This Business Object accumulates the responses as regards CDRs per EVSE by all CPOs.

ATTRIBUTE	DEFINITION	INST.	M/O	Format
RequestID	This is a unique identifier of the request, to which the CPO responds	1	M	string
CumulativeEnergyArea	This is the total energy delivered by all CPOs queried in the specific time period (in kWh)	1	M	double

8.5 PoDDemand

This Business objects models the energy demand per DSO fiscal smart meter with time. It can include historical data or forecasts.

ATTRIBUTE	DEFINITION	INST.	M/O	Format
PoDID	The unique identifier of the point of connection to the electricity grid.	1	M	string
EnergyDemand	List of energy transferred via the PoD with time (historical data or forecasts)	n	M	List of complex “EnergyTime”

9. Vehicle preparation Business Objects

This section describes models of the data structures, which are relevant for Business Scenario 7. This Business Scenario aims to plan the vehicle preconditioning, so that its battery capacity is maximum at the time of drive-off.

9.1 FleetMobilityNeed

This Business Object describes the data structure that is sent from the operator of a fleet of vehicles, in order to schedule charging of all vehicles and their optimum preparation based on their needs.

ATTRIBUTE	DEFINITION	INST.	M/O	Format
eMAID	This is the eMobility Account Identifier, as in ISO/IEC 15118. It refers to the account of an electromobility customer (with contract ID, service levels, etc.) and thus to any kind of an electromobility service that the service provider can offer.	1	M	string
RequestID	This is a unique identifier of the request, created by the receiver (EMP or NeMo)	1	M	string
UserMobilityNeed	This is a list of mobility needs per each vehicle in the fleet	n	M	List of complex “UserMobilityNeed”

9.2 VehiclePredictedEnergyNeed

This Business Object describes the data structure that predicts the energy need for the possible next trip of a specific vehicle.

ATTRIBUTE	DEFINITION	INST.	M/O	Format
Vehicle	This is a reference to the vehicle for which the prediction is done.	1	M	complex “Vehicle”
PredictedTripStart	This is the predicted time of start of the vehicle next trip	1	M	datetime
PredictedSoCNeedAtStart	This is the predicted need for battery SoC (in %) at the start of the predicted next trip of the vehicle	1	M	double
ProbabilityLevel	This is the probability level (in %) for the	1	M	double

	predictions as regards the start time of the vehicle next trip and the needed battery SoC at the start of the next trip.			

9.3 PreconditioningProfile

This Business Object describes the data structure including notifications for the preparation of vehicle functionalities, so that its battery is at the maximum SoC and the vehicle is optimally ready to start at the predicted time of the next trip.

ATTRIBUTE	DEFINITION	INST.	M/O	Format
Vehicle	This is a reference to the vehicle for which the prediction is done.	1	M	complex "Vehicle"
ChargingProfile	This is the proposed charging profile, to achieve the maximum battery SoC at the predicted trip start.	1	M	complex "ChargingProfile"
VehicleFunctionNotifications	This a list of suggestions on how to operate each vehicle functionality before the vehicle drive-off.	n	O	List of complex "VehicleFunctionNotification"

9.4 VehicleFunctionNotification

This Business Object describes proposals on how to operate vehicle functionalities before the vehicle drive-off.

ATTRIBUTE	DEFINITION	INST.	M/O	Format
Functionality	This is the vehicle functionality for which the proposal is made. Possible values are: 0 – "Heating" 1 – "AC" 2 – "Ventilator" 3 – "Seat heating" 4 – "Windshield defrost" 5 – "Odorant"	1	M	int
StartTime	This is the proposed time to turn on the functionality.	1	M	datetime
StopTime	This is the proposed time to turn off the functionality.	1	O	datetime

Level	This is the proposed level to set the functionality will be set. This is relevant for: 0 – “Heating” 1 – “AC” 2 – “Ventilator” 3 – “Seat heating” 4 – “Windshield defrost” 5 – “Odorant”	1	O	int
TemperatureSet	This is the proposed temperature to set the functionality. This is relevant for: 0 – “Heating” 1 – “AC” 3 – “Seat heating”	1	O	int

10. EV Charging infrastructure Business Objects

This section describes models of physical objects in the charging infrastructure of electric vehicles.

The Business Objects in this section have been based on the work done within the framework of the eMI^3 group, of which several NeMo partners are members. According to the eMI^3 specifications there are four levels of the charging infrastructure Business Objects: the Pool (Level 1), the Charging Station (Level 2), the EVSE (Level 3) and the Plug (Level 4).According to the specifications, the Pool is the Business Object used for search and find services.

The eMI^3 models have been updated according to the current views of the consortium members and the NeMo Hyper-Network needs.

10.1 ChargingConnector

This Business Object describes the connection of the vehicle charging cable with the electricity network. It can be a plug, socket or wireless connector, this is the name of the Object is ChargingConnector, while the corresponding Object in eMI^3 specifications was Connector / Socket-Outlet. The following description is based on the definitions of the Connector / Socket-Outlet object of the eMI^3 specifications, but extended so as to track real time information of a physical connector object.

ATTRIBUTE	DEFINITION	INST.	M/O	Format	Update of eMI^3 specifications
ChargingConnectorId	The global unique identifier of the Charging Connector.	1	M	string	This is a new attribute
Label	This is the identification label displayed on a socket outlet. It is the same as what is visible to a customer.	1	O	string	
TypeOfConnector	This is the type of the connector. Possible values are: 1-13: “DOMESTIC-A” to “DOMESTIC-M” 14: “IEC-62196-T2-F-NOCABLE” 15: “IEC-62196-T2-F-CABLE” 16: “IEC-62196-T3C-F-NOCABLE” 17: “IEC-62196-T1-M-CABLE” 18: “IEC-62196-T1-F-NOCABLE” 19: “IEC-309-2-1PH” 20: “IEC-309-2-3PH” 21: “CHADEMO”	1	M	int	New name of the TypeOfPlug attribute

	22: “IEC-62196-T3A-F” 23: “NEMA-5-20” 24: “TESLA-SPECIFIC” 25: “AVCON” 26: “LARGE-PADDLE” 27: “SMALL-PADDLE” 28: “IEC-62196-T2-COMBO” 29: “IEC-62196-T1-COMBO” 30: “CHINA-PART2” 31: “CHINA-PART3” 32: “BPLC-SPECIFIC” 33: “MARECHAL” 34: “IEC-309-2-DC”				
MaxPower	This is the maximum power (in W) that can be delivered by the connector during a charging session.	1	M	double	This is a new attribute
CableLength	The length (in mm) of the cable available at the Connector	1	O	int	This is a new attribute
LastUpdate	This is the timestamp when the charging connector description has been last updated	1	O	datetime	This is a new attribute

The MaxPower attribute has been added because it can be used to calculate charging time and to determine compatibility of the connector and vehicle.

The NumPhases, RatedVoltage, RatedPowerLevel, RealTimePower and GuaranteedMinimumPower attributes that are included in the eMI^3 specifications are not relevant for the ChargingConnector but for the EVSE, so they have been removed from the model.

10.2 EVSE

The EVSE (or Charging Point) is a system than can charge one EV at a time. Its main attributes are energy and electric characteristics. It is often associated with a parking place and may have one electricity meter.

An EVSE may have one or more Charging Connectors.

ATTRIBUTE	**DEFINITION**	**INST.**	**M/O**	**Format**	**Update of eMI^3 specifications**
EVSEID	This is the global unique identifier of the EVSE, as	1	M	string	

		proposed by eMI[3]				
Label		This is the identification label on the EVSE which is visible to the customer	1	O	string	This is a new attribute
LastUsedTime		This is the timestamp when an EV was successfully charged at the EVSE without any fault during the charging session.	1	O	datetime	This attribute was in the ChargingStation in eMI[3], but it is relevant for the EVSE
Charging Solutions	Charging Mode	This is the charging mode of the EVSE according to IEC-61851 charging modes. It can have the following values: 1 – “Mode1-AC-1p” 2 – “Mode1-AC-3p” 3 – “Mode2-AC-1p” 4 – “Mode2-AC-3p” 5 – “Mode3-AC-3p” 6 – “Mode4-DC” 7 – “Legacy-Inductive”	1	M	int	
	RatedPowerLevel	This is the rated power level (in W) that the EVSE is capable of delivering under normal operation conditions.	1	M	double	
VoltageRating		This is the rated voltage (in V) at which the EVSE operates under normal operating conditions.	1	O	int	
AdminState		This is the administrative state of the EVSE and the time of transition to the state	1	O	complex “AdminState”	
OperationalState		This is the operational state of the EVSE and the time of transition to the state	1	O	complex “OperationalState”	

ChargingConnectors	This is the list of charging connectors available on the EVSE	n	M	List of complex “ChargingConnector”	This is a new attribute
ParkingInfo	This includes information about the parking space associated with this EVSE.	1	O	complex “ParkingInfo”	This has moved from the ChargingStation, as the parking space is associated with an EVSE and not with a Charging Station
PoDID	This is the identifier of the point of delivery of this EVSE. It is the identifier assigned by the DSO for the point of connection of the EVSE to the electricity grid.	1	O	string	
EnergyMeterID	This is the identifier for main meter or sub-meters as assigned by the EVSE operator.	1	O	string	
LastUpdate	This is the timestamp when the EVSE description has been last updated	1	O	datetime	This is a new attribute

The NumPhases attribute of the Connector / Socket-Outlet object of eMI^3, which is not relevant for the ChargingConnector but for the EVSE, is not included as a separate attribute in the above model, because the information is already included in the ChargingMode attribute.

10.3 ChargingStation

A Charging Station is a physical object with a User Interface. It involves one or more EVSEs, sharing a common user identification interface.

ATTRIBUTE	**DEFINITION**	**INST**	**M/O**	**Format**	**Update of eMI^3 specifications**

StationID	This is the global unique identifier of the Charging Station	1	M	string	
Label	The label on the charging station, used to distinguish it among other stations.	1	O	string	
AuthorisationMethods	This is a list of methods how the user can become authorised to use and unlock the charging station. If under a contract with an EMP, the user can authenticate and then charge and pay through the EMP. The following values are possible: 1 - "Free" 2- "Private" 3 - "RFID-MIFARE" 4 - "RFID-DESFIRE" 5 - "RFID-Calypso" 6 - "PINPad" 7- "Phone-App" 8 - "Phone-NFC" 9 - "15118-PLC" 10 - "15118-OTA"	n	M	List of int	This attribute is part of the AuthorisationMethods attribute, which has been split in AuthorisationMethods and Payment Methods
PaymentMethods	This is a list of methods how the user can pay for the use of the charging station. When the user is not customer of an EMP, the user can pay directly. Possible values are: 1 - "Contract" 2 – “Phone” 3 – “Phone-SMS” 4 - "Credit-Card" 5 - "Cash" 6- "Pre-paid" Multiple values are possible	n	M	List of int	This attribute is part of the AuthorisationMethods attribute, which has been split in AuthorisationMethods and Payment Methods

UserInterfaceFeatures	These are the user interface features supported by the Charging Station. The following values are possible: 1 – “PinPadPresent 2 – “FeatureScreenPresent”	n	O	List of int	The Boolean values of the eMI[3] specifications have been replaced with a list of integers.
LanguageSupported	This is a list of languages supported by the user interface of the Charging Station. The languages are given as in IETF BCP-47 scheme (ISO 639 language codes+ ISO 3166 Country codes), for example en-US, en-GB, fr-FR, fr-CA	n	O	List of string	
LocationInfo	This is the location of the Charging Station.	1	M	complex “LocationInfo”	Improved definition of the attribute
Manufacturer	This is the manufacturer of the Charging Station	1	O	string	
Model	This is the model of the Charging Station	1	O	string	
StationType	This is the type of the Charging Station. Possible values are: 1- “Fixed” 2 - “Mobile”.	1	O	int	Forma has changed from text to integer
EVSE	This is the list of EVSEs belonging to the Charging Station	n	M	List of complex “EVSE”	This is a new attribute
AdminState	This is the administrative state of the charging station	1	O	complex “AdminState”	
UserComments	This is a list of comments about the Charging Station, possibly provided by users.	n	O	List of complex “UserComments”	

ServiceProviderComments	This is a description of the charging station as provided by the service provider	1	O	string	
LastUpdate	This is the timestamp when the Charging Station description has been last updated	1	O	datetime	This is a new attribute

The LastUsed and ParkingInfo attributes have been transferred to the EVSE model, as they are relevant for that and not for the Charging Station.

The OpenHours attribute has been transferred to the ChargingPool, as it is relevant for that object.

10.4 ChargingPool

This Business Object describes a location of one or more Charging Stations operated by one CPO.

ATTRIBUTE	**DEFINITION**	**INST**	**M/O**	**Format**	**Update of eMI[3] specifications**
PoolId	This is the global unique identifier of the Charging Pool.	1	M	string	
Name	This is the official name of the Charging Pool.	1	M	string	
Description	This is a free form textual description of a pool.	1	O	string	
LocationInfo	This is the location of the centre of the set of Charging Stations included in the Charging Pool.	1	M	complex “LocationInfo”	The definition has changed to point to the centre of the area
ChargingStations	This is the list of charging stations belonging to the pool	n	M	List of complex “ChargingStation”	This is a new attribute. It is necessary to know which charging stations belong to the

					pool
OpenHours	This is a list of time periods when the Charging Pool is open to the public.	n	O	List of complex “OpenHours”	
Accessibility	This indicates the accessibility of the Charging Pool. Possible values are: 1 – “Public” 2 – “Private” 3 – “Restricted” (for example for employees of a company) 4 – “Unknown”	1	O	int	This is a new attribute
TimeZoneOffset	This is the time zone of the Charging Pool location in ISO8601 UTC+Offset, i.e. [{+|-}hh:mm].	1	M	string	
TimeZoneName	This is the time zone name as in http://www.iana.org/time-zones/repository/tz-link.html	1	M	string	
CPO	This is the reference to the operator who operates the Charging Pool	1	M	complex “CPO”	This replaces the PoolContactInfo and OperatorContactInfo attributes to include a complete information about the operator via a new object
AdminState	This is the administrative state of the Charging Pool	1	M	complex “AdminState”	
LastUpdate	This is the timestamp when the Charging Pool description has been last updated	1	O	datetime	This is a new attribute

11. Support Objects

11.1 AddressInfo

This Support Object describes an address. The attributes were included in the LocationInfo object in the eMI^3 specifications. Since an address is needed in more objects and not only in the LocationInfo object, a separate Business Object for an address is proposed in the present document.

ATTRIBUTE	DEFINITION	INST.	M/O	Format	Update of eMI^3 specifications
StreetName	The Street Name (or road/ motorway number if no street name) where the entrance to the object is located.	1	M	string	
BuildingIdentifier	The building number (or name if it has no number) where the entrance to the object is located.	1	M	string	Updated definition to ensure flexibility when no number is available
BuildingIdentifier Addition	The building identifier addition where the entrance to the object is located. This can be additions to numbers like A, Bis, II etc., but should not contain street name, postal code or city information.	1	O	string	
Storey	This is the storey where the object is.	1	O	string	
PostalCode	The Postal Code of the area where the entrance to the object is located.	1	M	string	
PostalCodeAddition	The Postal Code addition (used in some countries) where the entrance to the object is located.	1	O	string	
City	The City/Town/Location where the entrance to the object is located.	1	M	string	

Country	The Country where the entrance to the object is located, following the ISO 3166-1 Alpha-2 Country Codes.	1	M	string	

The attributes AddresLine1 and AddressLine2 have been eliminated, as they are covered by the rest attributes.

11.2 AdminState

This Support Object contains information on whether a Business Object can provide service or not.

ATTRIBUTE	DEFINITION	INST.	M/O	Format	Update of eMI[3] specifications
AdminState	This is the administrative state of the object. It can have the following values: 1 – "Planned / In-deployment" 2 – "In-Maintenance" 3 – "In-Service" 4 – "Out-of-Service" 5 – "Deleted"	1	O	int	The format has changed to int
EventTime	This is the timestamp of transition to the indicated state	1	O	datetime	

11.3 Appointment

This Support Object models an appointment scheduled at a specific location and time.

ATTRIBUTE	DEFINITION	INST.	M/O	Format
User	This is a reference to a User	1	M	complex "User"
Date	This is the date when the appointment is scheduled	1	M	date
StartEndTime	Start and end time of the interval when appointment is scheduled	1	M	complex "TimePeriod"
Place	This is the location where the appointment will	1	M	complex

	be held			“LocationInfo”

11.4 Contact

This Support Object describes a contact person. The Contact and the CPO Business Objects combine and make more explicit and generic the proposed structures in section 2.1.1 of the eMI^3 specifications.

ATTRIBUTE	DEFINITION	INST.	M/O	Format	Update of eMI^3 specifications
personId	Identifier of the business partner person.	1	M	string	This is a new attribute
dn	Distinguished name of the user	1	O	string	This is a new attribute
Type	The person’s role in the partner. It can have the following values: 1 – “Main contact” 2 – “Commercial assistance” 3 – “Technical assistance” 4 – “Booking”	1	M	int	This is a new attribute
role	The person’s role in the Hyper-Network, if any. It can have the following values: 1 – “User” 2 – “Administrator” 3 – “Operator”	1	O	int	This is a new attribute
Surname	The last name of the person	1	M	string	This is a new attribute
MiddleName	The middle name of the person	1	O	string	This is a new attribute
FirstName	The first name of the person	1	M	string	This is a new attribute
Department	Department the user belongs to	1	O	string	This is a new attribute

AddressInfo	This is the address where the person is reachable	1	O	complex “AddressInfo”	This replaces all the relevant to address attributes in section 2.1.1
FixedPhone	This is the person’s direct phone, in international format	1	O	string	
MobilePhone	This is the person’s mobile phone, in international format	1	O	string	
Facsimile	This is the person’s facsimile phone, in international format	1	O	string	
Email	This is the person’s email	1	O	string	
Website	This is a website relevant to the person	1	O	string	
LanguageCode	This is a list of languages spoken by the person. The languages are given as in IETF BCP-47 scheme (ISO 639 language codes+ ISO 3166 Country codes), for example en-US, en-GB, fr-FR, fr-CA	n	O	List of string	
Preferred contact	This describes whether this is a preferred contact person. It can have the following values: 1 – “Yes” 2 – “No”	1	O	int	

11.5 CostOffer

This Support Object models the offered cost per kWh per day and time in an area.

ATTRIBUTE	DEFINITION	INST.	M/O	Format
FromTo	The start and end date and time of the time interval	1	M	complex “TimePeriod”

Cost	The cost per kWh in the time interval in an area	1	M	float
Currency	The currency in which the cost is expressed, according to ISO 4217 numeric code	1	M	int

11.6 ChargingProfile

This Support Object models the planning of a charging session. A charging session plan starts with the vehicle connection to the EVSE and ends with the vehicle disconnection. A charging session consists of one or more charging periods, the power at which energy is transferred is constant during a charging period.

Such a planning may be sent by the service provider to the CPO, including the requested charging profile. This object may also model the response by the CPO, as regards the offered charging profile.

ATTRIBUTE	DEFINITION	INST.	M/O.	Format
StartEnd	Start and date and time of the charging session	1	M	complex “TimePeriod”
Energy	The energy to be transferred to the vehicle during the charging period (in kWh)	1	M	double
ChargingProfilePeriods	The list of charging periods which will comprise the charging session	n	M	List of complex “ChargingProfilePeriod”
GreenEnergy	This indicates whether the energy offered comes from green sources. Possible values are: 1 – “Only green sources” 2 – “Partially green sources” 3 – “No green sources”	1	int	

11.7 ChargingProfilePeriod

This Support Object models the planning, in timing and power level, of one charging period. Each charging session consists of one or more charging periods.

ATTRIBUTE	DEFINITION	INST.	M/O.	Format
StartEnd	Start and end date and time of the charging period	1	M	complex “TimePeriod”
Power	The power (in W) at which energy is transferred during the period.	1	M	double

11.8 CPO

This Support Object describes the operator of a charging pool.

ATTRIBUTE	DEFINITION	INST	M/O	Format
CPOID	This is a global unique identifier of the CPO	1	M	string
CPOName	The name of the Charge Point Operator .	1	O	string
CPOContact	This is a list of contact persons for the CPO	n	M	List of complex “Contact”
ChargingPools	This is the list of charging pools that are operated by the CPO	n	M	List of complex “ChargingPool”

11.9 EnergyTime

This Support Object describes the maximum energy that is available in a specific time interval in a specific area.

It also models the energy demand in a specific area or per DSO fiscal meter in a specific time interval.

ATTRIBUTE	DEFINITION	INST.	M/O	Format
FromTo	The start and end date and time of each time interval	1	M	complex “TimePeriod”
Energy	The maximum energy (in kWh) that can be offered in the time interval or The energy demand in the time interval	1	M	double

11.10 GeoCoordinate

This Support Object describes the geo-coordinates of a location. The object is as in the eMI[3] specifications.

ATTRIBUTE	DEFINITION	INST.	M/O	Format
Latitude	The latitude of the location.	1	M	double

Longitude	The longitude of the location.	1	M	double
Altitude	The altitude of the location.	1	O	double
CoordinateSystem	The coordinate system at which the information is provided. Possible values are: 1 - "WGS84" 2 - "WGS72" 3 - "WGS44" 4 - "GRS80" 5 – “NAD83” 6 – “NAD27” 7 – “OSGB36” 8 – “ED50” 9 – “GDA94” 10 – “JGD2011” 11 – “KGD2002” 12 – “TWD67” 13 – “TWD97” 14 – “BJS54” 15 – “XAS80” 16 – “GCJ-02” 17 – “BD-09” 18 – “PZ-90.11” 19 – “GTRF” 20 – “CGCS2000” 21 – “CGS-2000” 22 – “ITRF” 23 – “Hong Kong Principal Datum”	1	O	int

11.11 ItinerarySection

This Support Object describes one section of an itinerary. One trip consists of consecutive itinerary sections. For planning, the list of the sections is calculated and provided by the navigation system.

ATTRIBUTE	DEFINITION	INST.	M/O	Format
SectionStartGeocoordinates	The geo-coordinates of the start location of the section	1	M	complex “GeoCoordinate”
SectionEndGeocoordinates	The geo-coordinates of the end location of the section	1	M	complex “GeoCoordinate”

SectionStartEndTime	The start and end date and time of the section	1	M	complex “TimePeriod”
TransportMode	This is the transport mode used to perform the itinerary section. Possible values are: 1- “Car” 2- “Walking” 3 – “Bike” 4 – “Motorbike” 5 – “Bus” 6 – “Tram” 7 – “Metro 8 – “Train” 9 – “Other” 10 - “Unknown”	1	M	int

11.12 LocationInfo

This Support Object describes a location.

ATTRIBUTE	DEFINITION	INST.	M/O	Format	Update of eMI^3 specifications
Name	This is the name of the location.	1	M	string	
Description	This is a description of the location.	1	O	string	
AddressInfo	This is the address of the location.	1	M	complex “AddressInfo”	
GeoCoordinate	The geo-coordinates of the centre of the location.	1	M	complex “GeoCoordinate”	
ParkingEntranceGeoCoordinate	The geo-coordinates of the vehicle entrance to the location.	1	O	complex “GeoCoordinate”	

The PropertyOwner attribute in eMI^3 specifications has been removed. This is because this object is proposed as a Support object that can be used to describe also vehicle locations and trips, so the location may not be always connected to an owner. If the location refers to a Charging Infrastructure object, then the Owner is included via the CPO object.

11.13 OpenHours

This Support Object describes the time periods when a Charging Pool or a Parking place is open to the public. This indicates the availability of a public charging station or the period when a private pool becomes a public one.

ATTRIBUTE	DEFINITION	INST.	M/O	Format	Update of eMI^3 specifications
AnnualClosings	This is a list of time intervals when the Charging Pool is closed.	n	O	List of complex "TimePeriod"	New format is proposed
RegularOpenings	This is a list of time intervals when the Charging Pool is open.	n	O	Map of Days ("Monday", "Tuesday", etc.) and complex "TimePeriods"	Improved definition and merged attributes

The Days and OpenHours attributes from eMI^3 have been eliminated, since they are included in the RegularOpenings.

The AnnualOpenings attribute is not used, since its difference from RegularOpenings is minor.

11.14 OperationalState

This Support Object describes the actual status of an EVSE when its administrative services is "In-service".

ATTRIBUTE	DEFINITION	INST.	M/O	Format	Update of eMI^3 specifications
OperState	This is the operational status of the EVSE. It can have the following values: 1 - "Unspecified" 2 - "Available" 3 - "Charging" 4 - "Faulted" 5 - "Unavailable" 6 - "Offline" 7 - "Reserved" 8 - "Other"	1	O	int	

	9 - “Unknown”				
EventTime	This is the time of transition to the indicated state	1	O	datetime	

11.15 ParkingInfo

This Support Object describes a parking space.

ATTRIBUTE	DEFINITION	INST.	M/O	Format	Update of eMI[3] specifications
SpotId	This is a global unique identifier for the parking spot.	1	M	string	
SpotStatus	This indicates the spot status. It can have the following values: 1 - “Free” 2 - “Occupied” 3 - “Reserved” 4 - “Not accessible”	1	O	int	New format proposed
SpotLabel	This is the identification label displayed on the parking spot, as seen by a customer.	1	O	string	
SpotDescription	This is a textual description of the parking spot.	1	O	string	
OpenHours	These are the time periods when the parking spot is available to the public.	1	O	complex “OpenHours”	
UseRestrictions	This indicates possible restrictions in using the parking spot. It can have the following values: 1 - “EV only” 2 - “Must be plugged in” 3 - “Disabled person only” 4 - “Commercial vehicles”	1	O	int	New format proposed

11.16 PowerTime

This Support Object models the maximum power available in a specific time interval.

ATTRIBUTE	DEFINITION	INST.	M/O	Format
FromTo	The start and end date and time of each time interval	1	M	complex “TimePeriod”
MaxPower	The maximum power (in kW) offered in the specific time interval	1	M	double

11.17 RecurrentUserPlace

This Support Object models a location visited recurrently by a User.

ATTRIBUTE	DEFINITION	INST.	M/O	Format
User	This is a reference to a User	1	M	complex “User”
Place	This is a reference to the place	1	M	complex “LocationInfo”
TimeFrequency	This is the frequency in which the place is visited	1	M	complex “TimeFrequency”

11.18 RecurrentUserTrip

This Support Object models a recurrent trip for a User.

ATTRIBUTE	DEFINITION	INST.	M/O	Format
User	This is a reference to a User	1	M	complex “User”
Trip	This is a reference to a trip	1	M	complex “Trip”
TimeFrequency	This is the frequency in which the place is visited	1	M	complex “TimeFrequency”

11.19 TimeFrequency

This Support Object models the frequency of an event that occurs periodically.

ATTRIBUTE	DEFINITION	INST.	M/O	Format
TimePeriod	Start and end time of the period	1	M	complex "TimePeriod"
WeeklyDay	Days of the week (1 = Monday, 2 = Tuesday,…) when the event usually occurs	n	M	List of int
MonthlyDays	Days (in 1 to 31) of the month when the event usually occurs	n	M	List of int
Months	Months (in 1 to 12) of the year when the event usually occurs	n	M	List of int

11.20 TimePeriod

This Support Object models a time period from a start to an end timestamp.

ATTRIBUTE	DEFINITION	INST.	M/O	Format
StartTime	This is the timestamp when the time period starts	1	M	datetime
EndTime	This is the timestamp when the time period ends	1	M	datetime

11.21 Trip

This Support Object models a trip.

ATTRIBUTE	DEFINITION	INST.	M/O	Format
TripStartEndTime	Start and end date and time of a trip	1	M	Complex "TimePeriod"
TripStartGeocoordinates	The geo-coordinates of the start location of the trip.	1	M	complex "GeoCoordinates"
TripEndGeocoordinates	The geo-coordinates of the end location of the trip.	1	M	complex "GeoCoordinates"

ItinerarySections	This is a list of itinerary sections that comprise the trip	n	M	List of complex “ItinerarySection”
LocationsVisited	This is a list of places visited by the user in the trip	n	O	List of complex “LocationInfo”

11.22 UserComments

This Support Object describes comments given by a user via the interface available at a Charging Station.

ATTRIBUTE	DEFINITION	INST.	M/O	Format	Update of eMI^3 specifications
StationID	This is the identifier of the charging station where the comment was given.	1	M	string	
Submitter	This is an identifier of the submitter of the comment, typically as a user handle within the namespace of the EVSP	1	O	string	
DateTime	The date and time when comment was received	1	O	datetime	
Comment	This is the content of the comment.	n	M	string	

12. Conclusions

Electromobility is currently impeded by the lack of interoperable systems and services. In this light, the NeMo project has proposed models for business objects and data structures in the electromobility sector. These Common Information Models describe objects relating to an electric vehicle, the charging infrastructure, the final user, the charge session, smart charging functionalities, the marketplace for service creation and delivery, reporting on grid loads due to the charging of electric vehicles and to the preparation of electric vehicles for drive-off. The work was based on the eMI³ specifications for the charge infrastructure objects and has extended to all other objects categories. The proposed Common Information models will form the basis for the service creation and provision within the NeMo Hyper-Network and to build translators so as to translate data from proprietary formats to the NeMo language. It is expected that this common language will promote the interoperability of electromobility services and will enable the better user's experience.

It is expected that during the project developments and the low level design within WP4, some changes will be necessary in the proposals presented in this document. In such cases, the proposed models will be updated accordingly.

References

eMi^3 Electric Vehicle ICT Interface Specifications Part 2 : Business Objects V1.0, 22/04/2015, available at: http://emi3group.com/documents-links/

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For more information:

NeMo Project Coordinator:

Dr. Angelos Amditis,

Research Director

Institute of Communication & Computer Systems (ICCS)

9 Iroon Politechniou Str.

GR-157 73 Zografou, Greece

a.amditis@iccs.gr

http://nemo-emobility.eu

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